

INTEGRATIVE BIOINFORMATICS ANALYSIS OF CONSTIPATION-ASSOCIATED GENETIC PATHWAYS AND THEIR OVERLAP WITH MENTAL ILLNESS.

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by

KUMARI KARUNA
(Roll No. -2624880005)

**UNDER THE SUPERVISION
OF
DR. PRITEE CHUNARKAR-PATIL**



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Declaration & Undertaking

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission.

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ABSTRACT

Psychiatric illnesses such as depression and anxiety often coincide with gastrointestinal issues, especially constipation. This indicates that a common biological factor might be involved, which is mediated by the gut-brain axis. The gut-brain axis is a sophisticated bi directional network which involve neural, endocrine, and immune signalling pathways that jointly regulate the central nervous system and the gastrointestinal tract. Although previous studies mainly looked at the role of microbiota and neural signalings, the role of the genetic host and molecular pathways in explaining the association between neuropsychiatric and gastrointestinal disorders has not been adequately examined.

In this research, gene sets related to constipation, depression, and anxiety were systematically collected from curated biomedical databases such as the Open Targets Platform and the Comparative Toxicogenomic Database (CTD), which compile evidence from genome-wide association studies, experiments, and literature curation. The lists of genes were pre-processed and standardized so as to be consistent and accurate across data sets. Intersection analysis was then done in order to identify the genes that are common to all three conditions. In total, 380 overlapping genes were found which may represent a core gene set responsible for the observed comorbidity of gastrointestinal and neuropsychiatric disorders.

In order to understand the biological importance of these shared genes, detailed functional enrichment analysis was carried out. Gene Ontology (GO) enrichment analysis, carried out under the Biological Process and Molecular Function categories, showed that the genes were significantly enriched in processes connected with intracellular signal transduction, regulation of gene expression, inflammatory response, and cytokine-mediated signalling pathways. Molecular function analysis revealed that the genes were enriched in receptor-ligand activity, cytokine activity, ion channel activity, and signalling receptor binding, which underlines their role in molecular communication and signal transduction mechanisms.

KEGG pathway enrichment analysis has identified several most important pathways that are significantly related to the overlapping gene set, including MAPK signalling, PI3K-Akt signalling, TNF signalling, neuroactive ligand-receptor interaction, and hormone signalling pathways. These pathways play essential roles in regulation of cellular stress responses, activation of immune system, neuronal signalling, and regulation of endocrine system. Their enrichment implies that dysregulation of these pathways could be responsible for both gastrointestinal dysfunction and neuropsychiatric conditions.

Besides, to check the physiological importance of the identified genes, tissue-specific expression analysis was performed using the GTEx database. It showed that the overlapping genes can be found in a wide variety of tissues especially in the brain and gastrointestinal tissues. A detailed analysis of the brain-gut-immune tissues showed that they have coordinated expression patterns, which further supports their functional role in gut-brain communication. In addition, correlation analysis of the average brain and gut expression levels revealed that they are significantly positively correlated ($r = 0.553$, $p = 7.84e-32$), which indicates that their transcriptional regulation is coordinated across these systems.

In addition, protein–protein interaction (PPI) network analysis that was carried out to identify functional relationships between overlapping genes, indicated that the network created was very densely interconnected. Genes such as MAPK1, AKT1, and TNF are the key hub genes in the network and function as regulators of signaling and inflammatory pathways. They are also likely to be the important mediators of the shared molecular mechanisms that are identified in this study.

Overall, the findings of this study provide strong evidence that constipation and psychiatric disorders are interconnected through shared genetic and molecular pathways involving neuroimmune, neuroendocrine, and signalling mechanisms. This study highlights the importance of host genetic factors in gut brain axis research and provides a systems level understanding of gastrointestinal neuropsychiatric comorbidity. The identification of shared genes and pathways offers potential insights into novel biomarkers and therapeutic targets. However, further experimental validation using transcriptomic and functional studies is required to confirm the biological roles of these findings.

KEYWORDS: Gut–Brain Axis, Constipation, Depression, Anxiety, Gene Overlap Analysis, Functional Enrichment Analysis, KEGG Pathway Analysis, Protein–Protein Interaction Network, MAPK Signalling.

CHAPTER 1

INTRODUCTION

1. Introduction

1.1 Background

Communication between the gastrointestinal tract and the brain occurs through a bidirectional network called the gut-brain axis. This network combines neural, hormonal, immune, and metabolic signalling pathways in order to align gut functioning with changes in emotions, cognition, and behaviour (Cryan et al., 2012). Neural direct communication e.g. via the vagus nerve and the enteric nervous system (ENS) is dominant but there are also other means of communication like hormones, cytokines, and microbial metabolites (Mayer et al., 2014). Normal physiological functioning depends on the uninterrupted communication between these two systems; however, disruptions in either system have been linked to a range of both gastrointestinal and neuropsychiatric disorders (Carabotti et al., 2015).

Constipation constitutes a large percentage of the global gastroenterology problem load and in addition to causing suffering it also negatively impacts the quality of life. Changes in the bowel movements characterize constipation. It is not simply a mechanical issue or one due to lifestyle factors. It has been shown to also be a neurobiological disorder with the potential of involving the ENS, also referred to as "second brain," (Furness, 2012) and the neurotransmitter signalling pathways it uses, to a great extent (Mawe & Hoffman, 2013). Additionally, it is not unusual for constipation to be associated with neuropsychiatric disorders like depression and anxiety (Adibi et al., 2022). This occurrence highlights common biological mechanisms that can explain the pattern of association rather than pointing to simple accidental overlap.

1.2 Constipation as a Neurobiological Disorder

The enteric nervous system (ENS) is a vastly complex and enormous network of nerve cells that is embedded in the gut walls. This is the main reason why it is called the 'brain of the gut' (Carabotti et al., 2015; Cryan & Dinan, 2012) in many cases. This sophisticated network of neurons is not only capable of sustaining life without any help from the brain but also it can work together with the central nervous system to regulate all the necessary functions of the digestive system such as intestinal movements, secretion, blood flow, and local immune responses. The ENS consists of neuronal circuits that are interconnected and primarily located in the myenteric and submucosal plexuses, and these two foldings of the gut are responsible for the optimal performance of the digestive functions (Dinan & Cryan, 2017).

Using neurotransmitters is essential to the roles of the ENS and the regulation of the intestinal motility. Serotonin (5-hydroxytryptamine), dopamine and acetylcholine in particular are major regulators of peristalsis and muscle contraction. Actually, the gut, especially the enterochromaffin cells, produces about 90% of the serotonin in the human body(Mawe & Hoffman, 2013). Such an observation strongly supports the importance of serotonin signalling in maintaining a healthy digestive system. Serotonin is related to the physiological processes of the gut such as secretion, sensory signalling, and coordinated muscle contractions leading to bowel movement.

Error in the ENS and neurotransmitter signalling can lead to huge changes in digestive motility. Problems such as impaired serotonin signalling, altered dopamine levels, or damaged cholinergic pathways can cause a reduction in peristaltic movements, deceleration of intestinal transit, and constipation(Mawe & Hoffman, 2013). These findings emphasize that constipation is not only the consequence of external factors like diet or lifestyle but is mainly attributed to neurophysiological abnormalities. Besides, GWAS are uncovering the genetic basis of gut motility disorders as they have identified gene variants associated with functions such as neuronal signalling, ion channel activity, and smooth muscle operation(Luo et al., 2025). Genes like those can influence the responsiveness of enteric neurons, the movement of electrolytes across the intestinal membranes, and the behaviour of intestinal smooth muscles in terms of contraction. Taken together, all these data advocate that constipation is a multifactorial disorder involving neurogenic and genetic factors and not simply a mechanical or behavioural problem.

1.3 Mental Illness and Gastrointestinal Dysfunction

Depression and anxiety are two of the most common neuropsychiatric disorders globally, which together make up a major health issue worldwide as they severely affect quality of life, productivity and general well-being(Miller & Raison, 2016; Smoller et al., 2019) They are complex disorders which may result from a mix of genetic, environmental, and psychosocial factors. Psychiatric genomics has shown that disorders of this nature are mostly polygenic, i.e. the influence of several hundred genetic variants each having a small effect is the major determining factor here, plus on top of that there are also environmental factors like stress, lifestyle, and early-life experiences acting as triggers(Miller & Raison, 2016; Smoller et al., 2019).

Besides, GWAS are revealing the genetic inheritance of gut motility disorders as they have pinpointed the gene variants associated with the biological functions like neuronal signalling,

ion channel activity, and smooth muscle performance(Luo et al., 2025). Such genes may determine enteric neuron reaction, electrolyte transport through the intestinal membranes, and intestinal smooth muscle contraction behaviour. Overall, this set of information promotes the view that constipation is a multifactorial disorder with both neurogenic and genetic aetiologies, rather than a mere mechanical or behavioural problem(Luo et al., 2025).

Even though the link has been validated by study results for a long time, it is only partly via the notion of psychological stress getting to the gut through a change in gut muscles motility, secretion and opening of the gut lining that one can account for it(Miller & Raison, 2016). On the contrary, recent research are revealing that the link may be the fact that the same or very similar neurobiological and molecular pathways become dysfunctional both in mental health disorders and gut conditions(Rogers et al., 2016). Neurotransmitter system dysfunction, immune system activation, inflammatory reactions, and neuroendocrine signalling changes are the main examples(Rogers et al., 2016).

The idea that both gastrointestinal motility disorders and psychiatric disorders can stem from one and the same biological mechanisms has been supported by the identification of common pathways(Rogers et al., 2016a) . These arguments do not contradict the concept of the gut brain axis which is a framework that describes the bidirectional interactions between the brain and the gut. In fact, it means that changes in one system can directly affect the other, and this is why constipation and mental health disorders are frequently found together(Carabotti et al., 2015; Cryan & Dinan, 2012).

1.4 The Gut–Brain Axis as a Connecting Framework

The gut brain axis is an idea which provides a biological explanation for the two-way interaction between gastrointestinal health and mental well-being(Carabotti et al., 2015; Cryan & Dinan, 2012). It is made up of the central nervous system and the nerves in the gut, the stress hormone system, the autonomic nervous system responsible for involuntary functions of the body, as well as the immune and chemical system of the body. These interconnected systems facilitate repeated mutual management of the gut and brain operations: vagal efferent signals influence gut movement, secretion, and mucosal immune system, whereas gut signals impact brain function, mood, and cognition.

Among immune messengers that circulate in the body, cytokines like TNF- α and interleukins are one of the ways through which these systems communicate with each other(Dinan & Cryan, 2017; Miller & Raison, 2016). They are capable of crossing the blood brain barrier or can alter

neuroinflammatory processes by stimulating endothelial receptors. Simultaneously, neuroendocrine signals such as cortisol produced via the activation of the HPA axis in response to psychological stress can have direct effects on increasing intestinal permeability, changing gut motility, and altering microbial composition(Dinan & Cryan, 2017; Miller & Raison, 2016).

While gut microbiota has been gaining a lot of attention as mediators of this kind of interaction(Mayer et al., 2014; Wang et al., 2023), host genetic factors affecting ENS development, neurotransmitter signalling, and GI motility have not been sufficiently studied, especially in relation to neuropsychiatric comorbidity(Luo et al., 2025). Constipation as a major symptom of ENS dysfunction constitutes a convenient model for studying the role of host genetic factors in gut–brain interactions(Mawe & Hoffman, 2013).

1.5 Current Limitations in Gut–Brain Axis Research

Many important gaps still persist in gut brain axis research despite some significant progresses(Cryan & Dinan, 2012; Rogers et al., 2016). The dominant view has been heavily oriented towards microbiota-driven mechanisms resulting in major discoveries of how microbial compositions and their metabolites modulate neurochemistry and immune regulation(Cryan & Dinan, 2012a; Mayer et al., 2014b). Nevertheless, this emphasis has marginalized the contribution of the host genetic makeup.

In particular, genes controlling the development and function of the ENS, neurotransmitter signalling pathways, and gastrointestinal motility have been overlooked in relation to neuropsychiatric disorders(Luo et al., 2025; Rogers et al., 2016). This problem is exacerbated by psychiatric researchers largely ignoring gastrointestinal symptoms such as constipation, and even regarding them as secondary manifestations or confounders rather than as biologically primary phenotypes worthy of separate investigation(Levy et al., 2006; Patil & Mehdi, 2025).

In method terms, the field has predominantly depended on clinical observations, epidemiological associations, and symptom-based correlations(Zhang et al., 2016). These methods, although they contain some valuable information, are not enough to reveal molecular mechanisms that link GI and psychiatric disorders. Complete, gene-oriented bioinformatics studies that are able to find mutual genes, signalling routes, and interaction networks are very infrequent(Luo et al., 2025; Rogers et al., 2016). The lack of such methods represents a major gap that holds back mechanistic knowledge as well as the discovery of potential therapeutic targets(Luo et al., 2025; Rogers et al., 2016).

1.6 Research Motivation and Objectives

It is a fact that constipation occurs alongside depression and anxiety very often, and this fact together with indicating that there are biological pathways in common lead to a gene-centric investigation of the molecular similarities between them. Some studies consider that constipation as one of the symptoms is a biologically separate phenotype though it is the one that shares the genetic and molecular structure of neuropsychiatric disorders through mechanisms of the gut–brain axis(Carabotti et al., 2015; Luo et al., 2025).

In the face of the critical points of the present studies, this piece of research is aimed at characterizing shared genetic loci and deregulated signalling pathways by applying a bioinformatics integrative approach.

Aim of the study:

- a) Find genes that are common to constipation, depression, and anxiety and are overlapping,
- b) Depict the biological processes and molecular mechanisms that are enriched,
- c) Check physiological relevance by performing tissue-specific expression analysis,
- d) Pick out hub genes as network regulators and therapeutic targets.

Such a system-level study will provide us with a deep insight into the molecular underpinnings of gastrointestinal neuropsychiatric comorbidity and pave the way for the future identification of a new set of biomarkers and therapeutic interventions(Cryan & Dinan, 2012).

CHAPTER 2

LITERATURE REVIEW

2. Literature Review

2.1 The Gut–Brain Axis as a Biological Communication Network

The gut-brain axis refers to a bidirectional communication network in which the gastrointestinal system and the central nervous system (CNS) interact and mutually influence their functions(Carabotti et al., 2015a; Cryan & Dinan, 2012b). Collectively, these systems coordinate physiological functions and maintain the body's internal environment stable(Carabotti et al., 2015a)

Initially, studies of the gut-brain axis were mainly focused on neural communication via the vagus nerve, which serves as the physical link between the brain and the gastrointestinal tract(Margolis et al., 2021)Furthermore, interaction between these two organs takes place at the molecular level with cell signaling molecules in the bloodstream such as hormones, cytokines, and neurotransmitters(Dinan et al., 2017). It is due to these signaling molecules that the gut and brain are continuously in feedback communication with one another(Carabotti et al., 2015; Cryan & Dinan, 2012).

Based on (Carabotti et al., 2015)study, the human gut has been recognized as a source of signals through which cognition, emotions, and behavior can be affected. In addition, brain-derived signals can regulate the gut's movements, secretion, and overall gastrointestinal physiology. This form of mutual communication not only displays the functionally integrated nature of these two systems but also suggests that they may not be as separate as we thought before(Carabotti et al., 2015).

On top of that,(Cryan et al., 2012)expanded the concept and stated that the gut-brain axis is not simply a channel of communication, but it plays a key role in the maintenance of physiological homeostasis. Malfunction within this axis has been linked, not only to digestive system disorders but also to mental illnesses (Cryan & Dinan, 2012). This means that if one system fails, the other system will be affected immediately. Therefore gut and brain health are linked at a very deep level(Cryan et al., 2012).

2.2 Enteric Nervous System and Neurotransmitter Regulation of Gut Motility

Among the many roles that the enteric nervous system- also known as the “second brain” (Furness, 2012)carries out in the human body, one stands out- that is its regulation of gastrointestinal motility(Carabotti et al., 2015; Mawe & Hoffman, 2013). Though functioning

quite independently, ENS has the ability to communicate with CNS through the gut-brain axis(Cryan & Dinan, 2012a). Additionally, ENS is responsible for secretion, blood flow, as well as, local immune responses of the gut(Carabotti et al., 2015b).

There is frequent use of neurotransmitters by the ENS to carry out various functions as one of its hallmarks. Serotonin, dopamine, acetylcholine are examples of neurotransmitters responsible for the regulation of motility and sensory signaling in the intestines(Mawe & Hoffman, 2013). Of all these neurotransmitters, serotonin specifically is considered a major player in the gut with the majority around 70-90% of the body's serotonin production taking place in the gastrointestinal tract(Mawe & Hoffman, 2013).

Serotonin plays a central role in regulating motility by coordinating peristaltic movement and smooth muscular contractions that allow the food to move along the intestinal tract(Mawe & Hoffman, 2013). A change in the serotonin signalling pathway might cause issues in the gut motility leading to delayed transit time and constipation(Mawe & Hoffman, 2013).

Imbalance in the neurotransmitters along with disruption in the cell signalling mechanisms of the ENS are the primary factors that result in the appearance of GI problems like chronic constipation(Mawe & Hoffman, 2013). Medically speaking, these findings suggest that constipation is more than just a physical problem; it is a neurogenic disorder that is linked to defective neural pathways. This idea unites two apparently unrelated areas, namely, gastrointestinal function and neural control(Mawe & Hoffman, 2013).

2.3 Genetic Basis of Constipation and Gastrointestinal Motility Disorders

Initially, constipation was blamed to a great extent on lifestyle factors such as insufficient fiber intake, dehydration, and lack of physical activity(Luo et al., 2025). But now, genetics has been recognized as an equally significant factor in the development of gastrointestinal motility disorders, thanks to genetic studies.

By using genetics, researchers have observed several gene mutations that affect nerve signaling, muscle movements, and ion channels in the digestive system, all of which play crucial roles in digestive motility(Luo et al., 2025). Therefore, gut motility seems to be regulated by intricate genetic networks rather than being solely subject to environmental influences(Luo et al., 2025).

Besides that, mutant genes influencing the nervous system, neurotransmitter production, and the balance of neurotransmitters have also been linked to the constipation problem(Luo et al.,

2025; Mawe & Hoffman, 2013). Thus, neurogenetic involvement in this health condition has been revealed. Thus, upon realizing that constipation is likely a genetic and neurobiological disorder rather than a functional or lifestyle-related problem, scientists have altered their perspective on the issue(Luo et al., 2025).

2.4 Mental Disorders and Gastrointestinal Dysfunction

Mental illnesses such as depression and anxiety arise from a complex interplay of hereditary and environmental factors(Miller & Raison, 2016b; Smoller et al., 2019b). They have the potential to change various biological systems, for example, neurotransmitter cell signaling, the immune system, and stress-related pathways.

From genetic research, it has become evident that the genes that predispose to mental disorders are those that regulate synaptic transmission, neurotransmitters, inflammatory responses, and cellular stress signals (Miller & Raison, 2016b; Rogers et al., 2016b). Given that those pathways are also implicated in the function of the gastrointestinal system, it can be concluded that the molecular basis of both mental and digestive disorders might be the same(Liu et al., 2024a; Rogers et al., 2016b).

Therefore, through investigations and clinical assessments, it turns out that individuals with depression and anxiety are highly likely to develop gastrointestinal symptoms such as constipation(Adibi et al., 2022; Ballou et al., 2019; Zhang et al., 2016). This demonstrates the tight and two-way relationship between mental health and gastrointestinal function(Adibi et al., 2022; Ballou et al., 2019).

Sharing biological pathways and symptoms makes a strong case for the idea that psychiatric and gastrointestinal disorders are not separate entities but rather two aspects of the same coin (Liu et al., 2024; Rogers et al., 2016).

2.5 Gut–Brain Axis as a Framework for Gastrointestinal–Neuropsychiatric Comorbidity

The gut–brain axis is a helpful illustration to clarify why gastrointestinal and neuropsychiatric disorders frequently coexist(Carabotti et al., 2015; Cryan & Dinan, 2012). It integrates the nervous system, hormonal system, immune system, and molecular signaling to demonstrate how a dysfunction in one can lead to problems in the other(Carabotti et al., 2015; Cryan & Dinan, 2012).

Brain diseases and gastrointestinal problems share biological mechanisms such as neurotransmitter signaling, immune cell activation, and stress-related pathways(Liu et al., 2024; Rogers et al., 2016). These common pathways suggest that both types of disorders may stem from similar molecular and physiological alterations.

Due to the bidirectional nature of the gut–brain axis, gut problems may influence brain functions, and the brain may also affect the gut(Cryan & Dinan, 2012; Margolis et al., 2021). For example, gut inflammation or decreased motility may alter mood and cognitive functions, while psychological stress may induce alterations in gut functions(Carabotti et al., 2015; Margolis et al., 2021).

This perspective on these diseases illuminates the fact that we should not consider them individually; rather, we should study them jointly since they are connected through common biological systems(Cryan & Dinan, 2012; Rogers et al., 2016).

2.6 Limitations of Existing Research and Identified Research Gap

There is still a number of issues with current gut–brain axis studies which have to be addressed(Cryan & Dinan, 2012; Rogers et al., 2016). For example, symptom-level associations or the role of gut microbiota are most frequently studied, whereas the importance of host genetic factors is often neglected.

Although microbiome research has led to new understanding, it fails to explain the molecular aspect of gastrointestinal–neuropsychiatric comorbidity fully(Mayer et al., 2014). Moreover, many researchers consider gastrointestinal symptoms, e.g., constipation, as incidental rather than as primary conditions(Levy et al., 2006; Patil & Mehdi, 2025).

Integrative, gene-centric bioinformatics studies to identify the genetic pathways common to gastrointestinal and mental disorders are scarce(Luo et al., 2025; Rogers et al., 2016). This lack of research impedes the comprehension of the fundamental molecular processes and the discovery of biomarkers and drug targets(Luo et al., 2025; Rogers et al., 2016).

This paper fills the void by using a bioinformatics approach to detect common genes and molecular pathways among constipation, depression, and anxiety(Luo et al., 2025). Concentrating on host genetic factors, this paper seeks to shed more light on the gut–brain axis's molecular nature(Carabotti et al., 2015; Luo et al., 2025).

CHAPTER 3

METHODOLOGY

3. Methodology

3.1 Study Design

This study was an integrative bioinformatics investigation using a cross-disease computational approach to analyse shared genetic and molecular mechanisms of constipation, depression, and anxiety in the gut-brain axis concept. Because of the strong clinical link among these conditions and the fact that their molecular basis is still not well understood, the use of systems biology focusing on genes was the best way to uncover the inter-disease genetic overlaps and their functional implications.

In the comparative multi-disease framework, genes associated with diseases of each concerned condition were first separately identified, standardized in format, and the overlap of their sets was analyzed in order to generate a core suite of common genetic elements. The motivation here was that although phenotypically different and clinically comorbid disorders may share molecular pathways, pointing out these pathways can be of considerable mechanistic importance for the study of co-morbidity between the digestive system and neuropsychiatric disorders.



Figure 1: Workflow of the study

3.2 Retrieval of Disease-Associated Gene Sets

Genes linked to diseases like constipation, depression, and anxiety were sourced from high-quality biomedical databases such as the Open Targets Platform and Comparative Toxicogenomics Database (CTD)(Davis et al., 2021; Ochoa et al., 2021). These resources compile data from GWAS, lab experiments, clinical studies, and curation of the literature to ensure a high level of confidence in gene-disease associations.

Gene lists for each disorder were acquired separately to get an unbiased and accurate picture of the genetic basis of each condition. Utilizing multiple databases helped to make the gene sets for this study more comprehensive and dependable.

3.3 Data Preprocessing and Gene Standardization

The gene lists that were obtained underwent preprocessing as one way of really making different datasets comparable and accurate. To avoid any discrepancies caused by synonyms, aliases, or old identifiers, gene symbols were standardized to the newest and most accepted naming conventions.

To keep things tidy, duplicate records were taken out, and non-standard or unidentified gene indicators were also excluded. This step turned out to be very important for the uniformity of the datasets. Besides allowing for a good comparison, it also made it possible to merge datasets properly in the downstream processes.

3.4 Identification of Overlapping Genes

To get to the bottom of shared genetic factors between constipation, depression, and anxiety, a gene set intersection analysis was undertaken. Genes that were common to all three sets were taken out to form a final overlapping gene pool.

This overlapping gene pool is a symptom of what molecular networks could be playing a part in the co-existence of gastrointestinal and neuropsychiatric conditions. The amount of overlap was most simply shown using a Venn diagram, which also served to illustrate the distribution of unique genes for the three conditions.

3.5 Functional Enrichment Analysis

3.5.1 Gene Ontology (GO) Enrichment Analysis

The roles that the overlapping genes play biologically were determined through Gene Ontology enrichment analysis (Carbon et al., 2021). Both Biological Process (BP) and Molecular Function (MF) were considered during the analysis. Biological Process was the category that pointed out different cellular functions like signaling inside the cell, the immune system's activity or even gene expression, whereas Molecular Function provided a description of the cells' biochemical activities - e.g. receptor binding, cytokine activity, or ion channel function.

To find GO terms significantly present at a higher level than you would expect by chance, a statistical overrepresentation analysis was used. To keep the number of false discoveries down, multiple testing corrections were made, and GO terms having a FDR-adjusted p-value less than 0.05 were considered to be statistically significant.

3.5.2 KEGG Pathway Enrichment Analysis

To find which biological pathways were most relevant to the gene overlap, KEGG pathway enrichment analysis was carried out (Kanehisa et al., 2021). The KEGG database is a great source of well-maintained molecular interaction networks such as signaling and pathways related to diseases.

Once the genes overlapping each condition were identified, they were placed on KEGG pathways so their roles in the biological systems might be discovered. After pathways came up with an adjusted p-value of less than 0.05, only those were accepted as the ones that significantly contributed to the study and were then arranged according to their degree of enrichment.

3.6 Data Visualization

Graphs were made for enrichment results using R language. The topmost enriched GO terms and KEGG pathways were represented using bar and dot plots respectively.

On bar plots, only enriched terms based on statistical significance were kept while dot plots went deeper and used both gene count and p-value to give a full picture of the enrichment results. Such plots have significantly aided the recognition of major biological processes and pathways.

3.7 Tissue-Specific Expression Analysis

To analyze at which tissue the overlapping genes might play a role, their expression data were retrieved from the GTEx (Genotype-Tissue Expression) database(Aguet et al., 2020). GTEx offers transcriptomics data of several human tissues, thus it is a useful resource for studying tissue-specific gene expression patterns.

We used median expression levels (TPM: Transcripts Per Million) to measure gene expression in different tissues. To understand the gene expression variations, we mainly looked at brain and gastrointestinal tissues. Gene expression heatmaps visually represented tissue-specific trends.

Along with the above-mentioned tissues, immune tissues were also used for analysis to support the idea of the gut–brain axis.

3.8 Correlation Analysis

The aim of this work was to figure out the degree of association between gene expression in brain and gastrointestinal tissues; therefore, a correlation study was carried out.

First, average expression values were computed for brain and gut tissues values separately per gene, then these were log₂-transformed to standardize the data and lessen the variability.

In Pearson correlation analysis, the relationship between two continuous variables is assessed. The analysis findings were also represented as a scatter plot that featured the best-fit regression line.

3.9 Protein–Protein Interaction (PPI) Network Analysis

We performed a Protein–protein interaction (PPI) network analysis to help us understand the functional interactions of the overlapping genes better.

One of the great features of the STRING database is it produces protein interaction networks by compiling known experimental interactions, computational predictions as well as interactions extracted from the literature(Szklarczyk et al., 2023). Based on the number of nodes that each gene was connected to, we could pinpoint the hub genes. The ones that connect with most neighbors are usually the most important and therefore, these hub genes were isolated for follow-up studies.

3.10 Statistical Criteria

All enrichment analyses were carried out using a cut-off level of statistical significance (adjusted p-value < 0.05). Multiple testing correction methods have been used to control the false discovery rate (FDR).

As for correlation analysis, the significance of the association between variables is evaluated by the p-value corresponding to the Pearson correlation coefficient.

CHAPTER 4

RESULT & DISCUSSION

4. Results

4.1 Gene Collection and Overlap Analysis

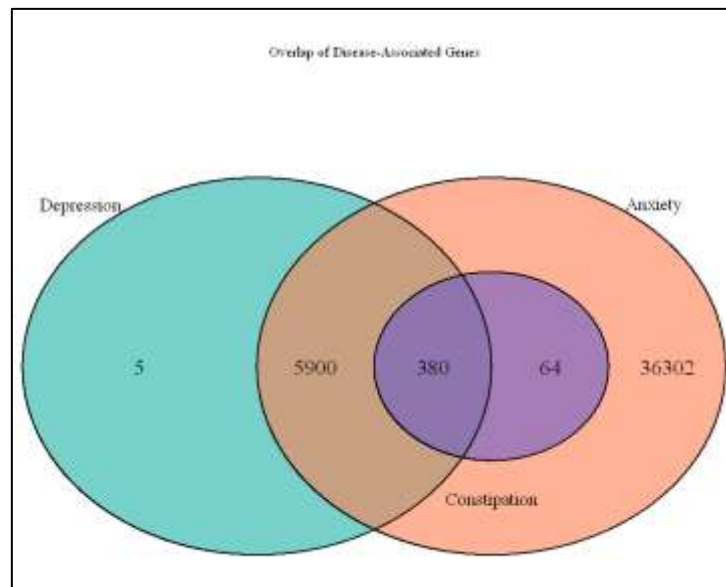


Figure 2: Venn diagram showing the overlap of genes associated with constipation, depression, and anxiety

A set of 380 genes were identified common between all the three disorders constipation, depression and anxiety

4.2.1 Biological Process Enrichment

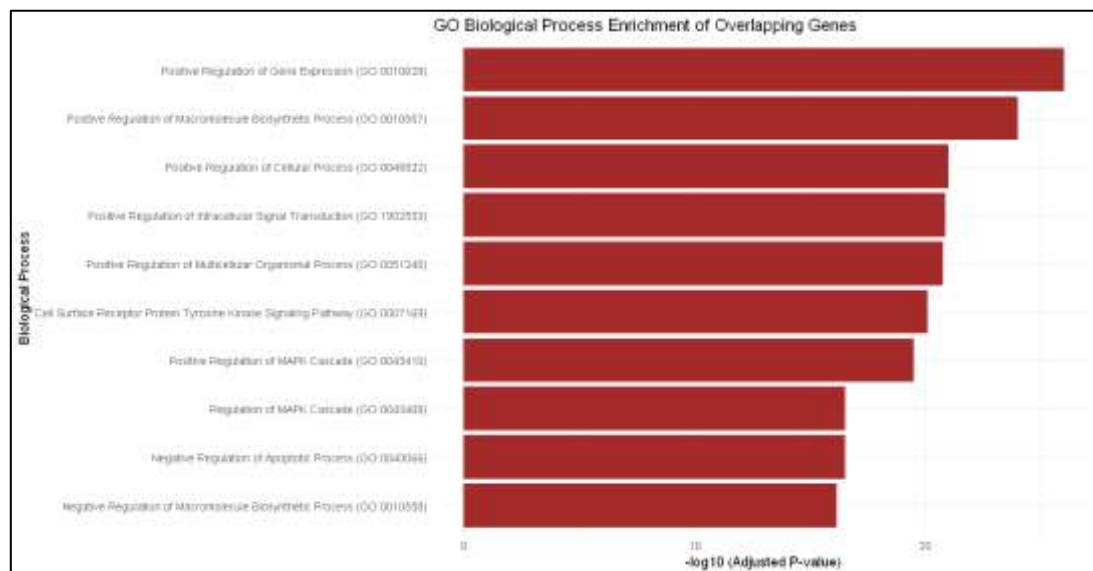


Figure 3 : Gene Ontology (Biological Process) enrichment analysis of overlapping genes.

The overlapping genes were significantly enriched in biological processes related to intracellular signaling, gene regulation, and inflammatory response.

4.2.2 Molecular Function Enrichment

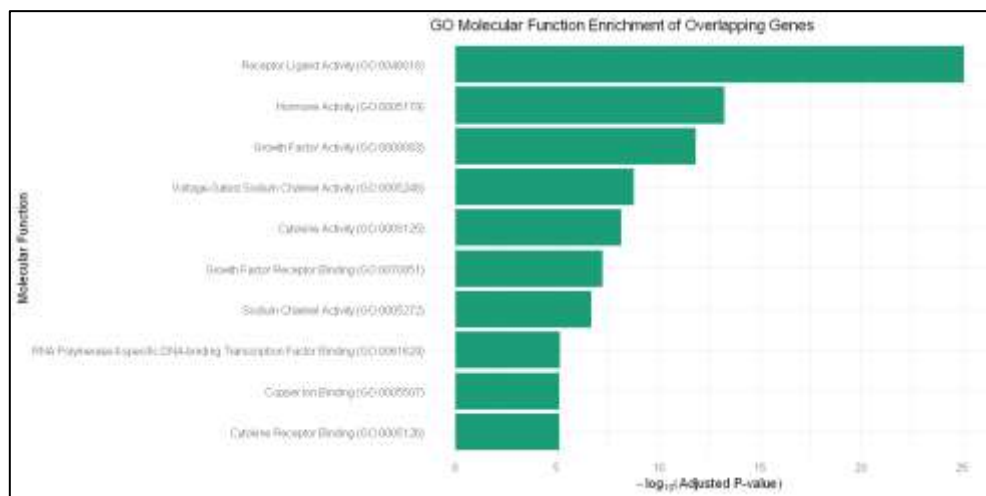


Figure 4:Gene Ontology (Molecular Function) enrichment analysis.

Molecular function analysis revealed enrichment in receptor–ligand activity, cytokine activity, and signaling receptor binding.

4.2.3 KEGG Pathway Enrichment Analysis

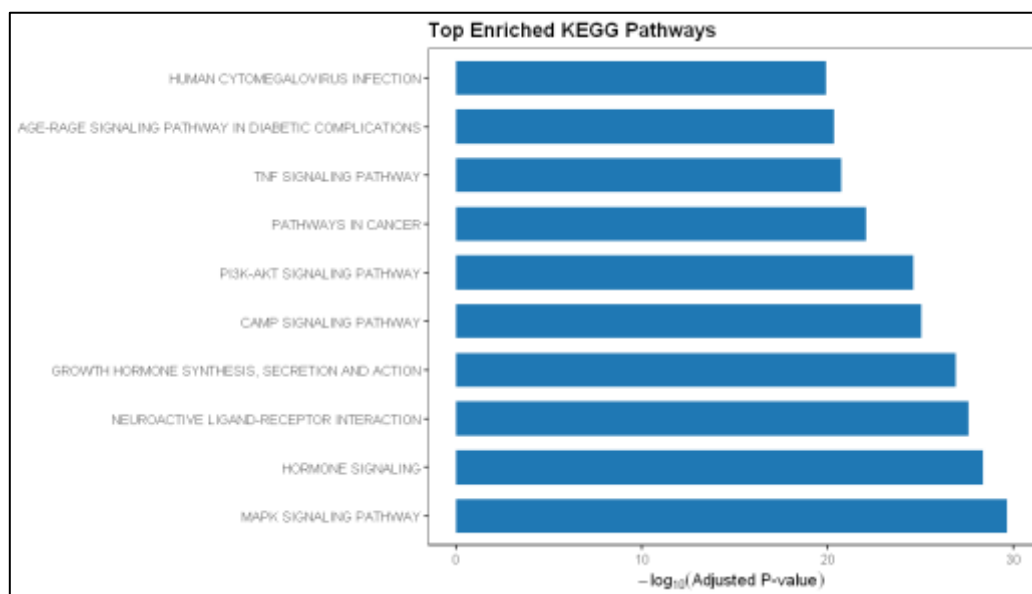


Figure 5:KEGG pathway enrichment analysis of overlapping genes highlighting significant pathways.

4.3 Tissue-Specific Expression Analysis

To explore the tissue level importance of the overlapping genes, expression data from the GTEX database was analysed

4.3.1 Global Tissue Expression Profile

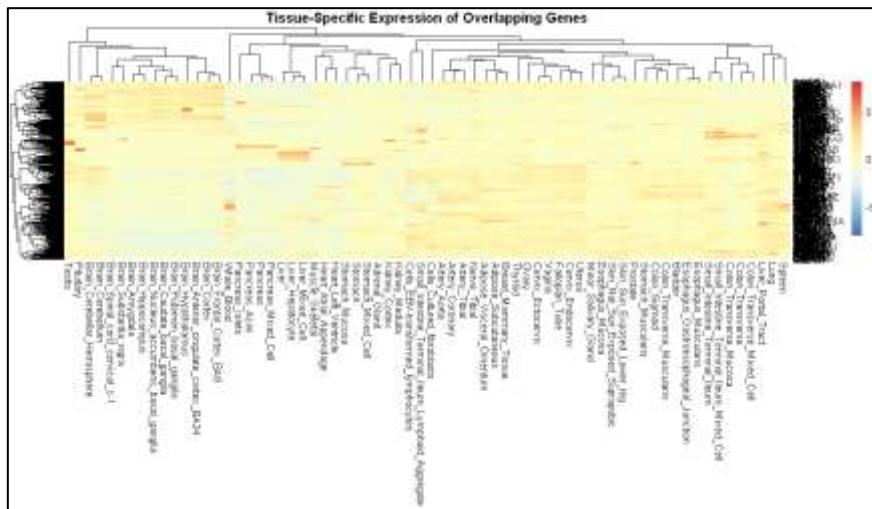


Figure 6: Heatmap showing global tissue-specific expression of overlapping genes across multiple human tissues.

A heatmap of gene expression across multiple tissues revealed that the overlapping genes are expressed in a broad range of tissues, with strong expression in brain and gut tissues as the major ones (Figure 7).

4.3.2 Focused Brain–Gut–Immune Expression Pattern

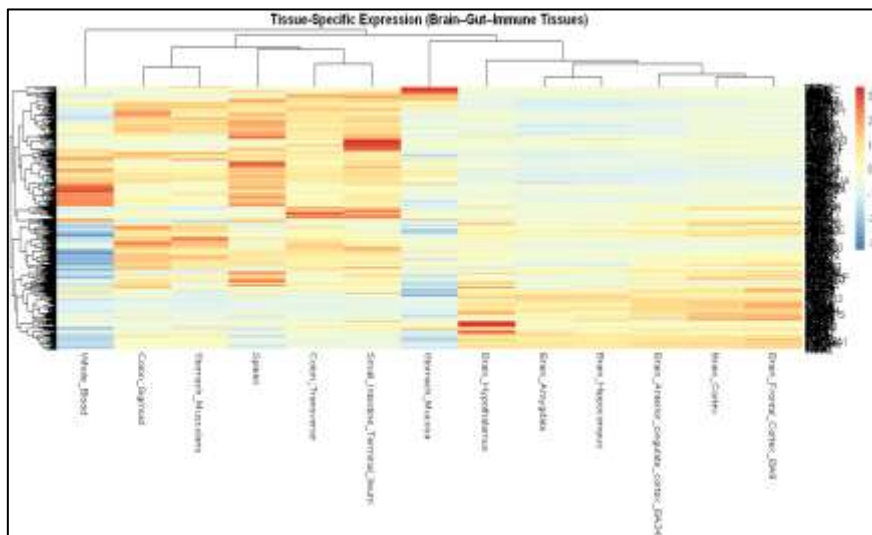


Figure 7: Heatmap representing focused expression patterns of overlapping genes in brain, gastrointestinal, and immune tissues.

Further dissecting gene expression in brain, gastrointestinal tract and immune tissues only, the results show very different patterns of gene expression in these tissues, suggesting that these genes are strongly expressed in the main players of the gut–brain axis (Figure 8).

4.3.3 Identification of Highly Expressed Genes

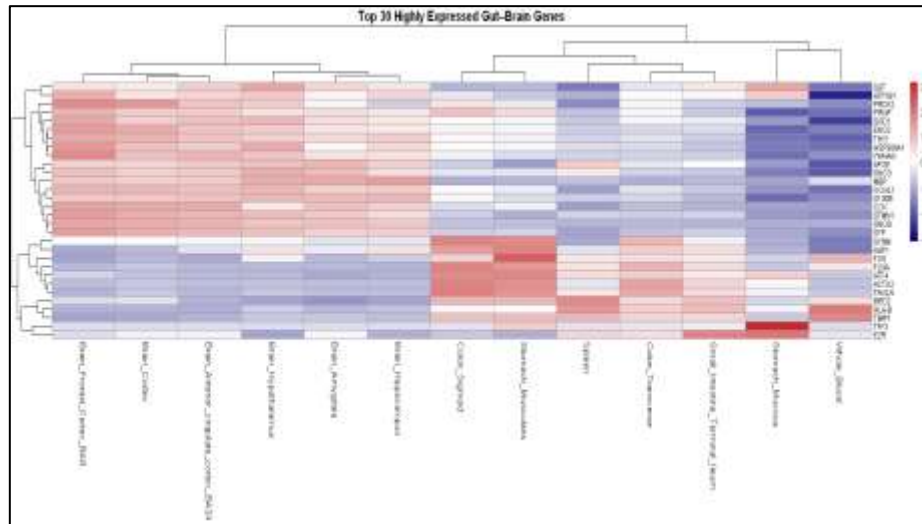


Figure 8: Heatmap of highly expressed genes in brain and gut tissues demonstrating coordinated expression patterns.

Some genes that are strongly expressed in the brain as well as gut tissues were selected on the basis of TPM values. Heatmap presentation of these genes indicated concerted expression patterns, thus implicating their possible function in gut–brain communication (Figure 9).

4.4 Correlation Analysis Between Brain and Gut Expression

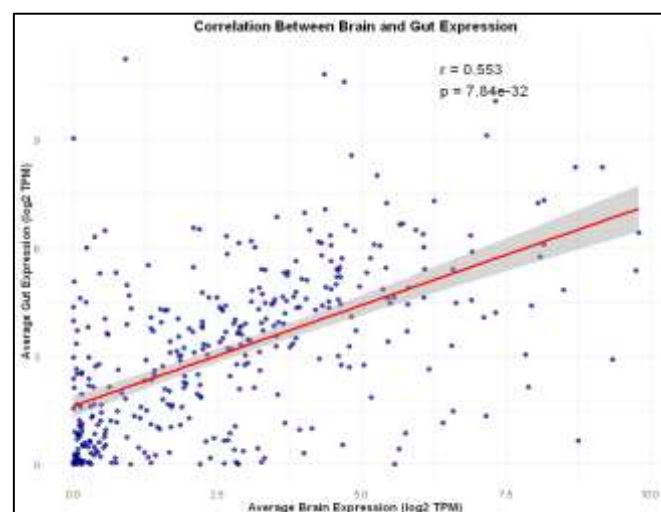


Figure 9: Scatter plot showing correlation between gene expression levels in brain and gastrointestinal tissues ($r = 0.553$), indicating coordinated transcriptional regulation.

A correlation analysis was performed to assess the relationship between gene expression in brain and gastrointestinal (GI) tissues.

Average brain and gut expression levels of the overlapping genes were positively correlated ($r = 0.553$, $p = 7.84e-32$), which suggests coordinated transcriptional regulation across these two tissues (Figure 8).

4.5 Protein–Protein Interaction (PPI) Network Analysis

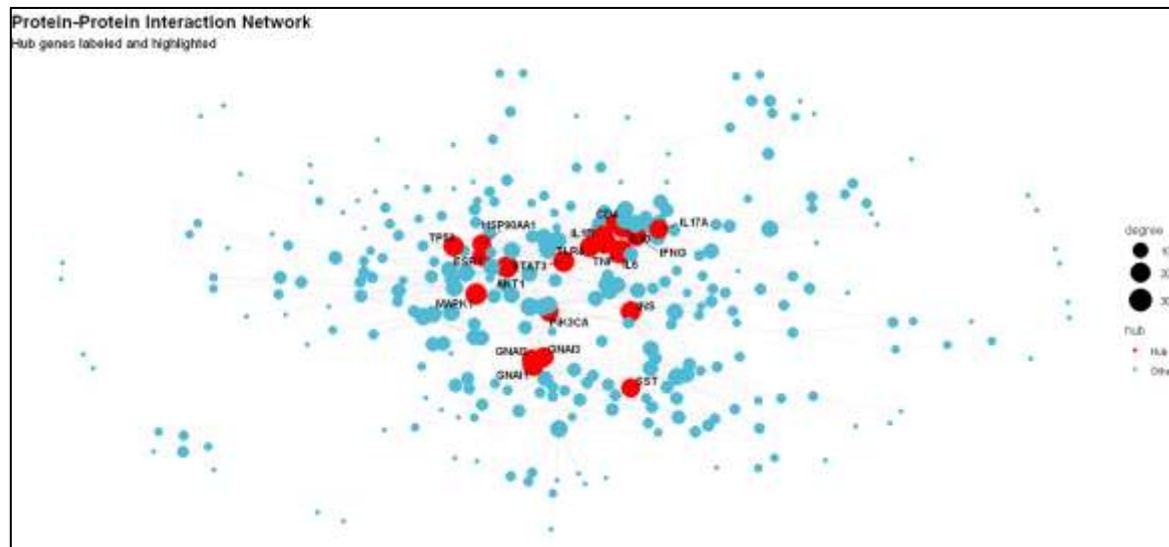


Figure 10:Protein–protein interaction (PPI) network of overlapping genes.

Protein-protein interaction (PPI) network analysis of the overlapping genes was conducted to identify key regulatory genes.

In the network, a number of very highly connected hub genes were found, such as:

- MAPK1
- AKT1
- TNF

Overall, these hub genes hold central positions in the network, implying their potential role as major regulators in the common signaling pathways between gastrointestinal and neuropsychiatric disorders (Figure 11).

DISCUSSION

5.1 Overview of Findings

This research uncovered 380 genes shared not only by constipation but also by depression and anxiety, which suggest that these conditions have a common genetic basis. The identified genes are mainly involved in intracellular communication, gene expression control, and immune functions, as determined by functional enrichment analysis. Significant pathways included MAPK, PI3K–Akt, TNF signalling, and neuroactive ligand–receptor interaction. Moreover, tissue-specific expression analysis showed that these genes are actively expressed in both the nervous and digestive systems, and their expression levels are significantly positively correlated ($r = 0.553$, $p = 7.84e-32$). Besides that, network analysis revealed important central hub genes, thereby highlighting the main regulatory features of this shared molecular network.

5.2 Shared Genetic Architecture of Gastrointestinal and Psychiatric Disorders

The fact that we found 380 genes common to constipation, depression, and anxiety is very strong evidence that a set of genes underpins these diseases in a similar way. Gastroenterologists and psychiatrists would, in most cases, think of constipation and depression/anxiety as two completely separate entities. Nevertheless, this new evidence of a vast number of shared genes suggests that these disorders come from biological systems that few have kept their doors closed to each other. Hence, the new findings verify the idea that one molecular framework could be at the basis of gastrointestinal and neuropsychiatric comorbidities.

5.3 Role of Intracellular Signalling Pathways

The gene set related to common biological processes, such as positive and negative regulation of signalling and response to stress, includes the MAPK pathway that was identified as one of the most significantly enriched ones among the common genes. Properties of this MAPK signalling pathway involves a series of biochemical reactions in the cell that lead to changes in the cell function including neurotransmitter release and cell death i.e. apoptosis. In addition to changes occurring at the cellular level in brain neurons descending inputs from the central nervous system to the enteric nervous system are regulated by MAPK as well. This could speculate on how alterations in pathways of this type have been implicated not only in brain function but also in motility disorders of the gut.

On the other hand, the PI3K-Akt pathway, which is known to be involved in regulation of apoptosis, metabolism, and cell growth, has a high level of the signal from this pathway or/and its constituent genes in the brain. The fact that this pathway is common to the diseases in question again points to the disruption of the regulation of cell death and survival mechanisms as the major underlying reason of both gut disorders and mental disorders.

5.4 Neuroimmune and Inflammatory Mechanisms

The association between neuroimmune and inflammatory processes with psychiatric as well as gastrointestinal disorders is supported by the results showing enrichment of cytokine activity and TNF signalling along with immune-related processes. The TNF signalling pathway is a major pathway involved in systemic inflammation and is also implicated in depression and the gastrointestinal disorders.

Changes in brain function caused by signals from cytokines can be a result of either the passage of the cytokines themselves through the blood-brain-barrier or the activation of receptors on endothelial cells leading to the release of secondary messengers capable of transmitting the signal into the brain. In the same time lesional gut inflammation can hamper the gut barrier function resulting in altered gut motility and constipation. In this way by the enrichment of immune-related pathways neuroimmune mechanisms underlying comorbidity between gut and brain function can be better understood.

5.5 Neurotransmission and Receptor-Mediated Signaling

From molecular function analysis we found that receptor-ligand activity, cytokine activity, and ion channel activity were particularly enriched, which indicates that the effective signaling is crucial not only at the cellular but also at the systemic level. The fact that neuroactive ligand-receptor interaction pathways were enriched neuroactive ligand–receptor interaction pathways further showcases the significance of neurotransmission for gut–brain tie.

Neurotransmitters like serotonin and dopamine and other neuroactive substances play a very important role in influencing mood as well as ensuring the proper functioning of the gastrointestinal tract. Besides that, signaling through receptors is implicated in the simultaneous occurrence of constipation and psychiatric symptoms. This is because the two disorders stem from the disruption of the same signal transduction pathways that have effects on both the brain and the gut at the same time.

5.6 Tissue-Level Coordination and Gut–Brain Connectivity

The expression analysis targeting specific tissues indicated that the common genes are expressed in numerous tissues while the highest levels of expression were observed in both the brain and gastrointestinal systems. Scrutiny of gene expression in the brain, gut, and immune tissues enabled the identification of expression profiles that were well-organized and led to the inference of functional integration of the three systems.

In fact, the correlation analysis showed a markedly strong positive correlation ($r = 0.553$, $p = 7.84 \times 10^{-32}$) between the gene expression levels of brain and gut tissues. This finding is very suggestive of the concurrent regulation of transcription and at the same time very encouraging to the notion that molecular connections exist between the central nervous system and the gastrointestinal tract. The gut–brain axis features as one of the main characteristics to the coordinated transcriptional activity of these genes.

5.7 Central Role of Hub Genes in Network Regulation

By means of protein-protein interaction network analysis, we identified some key hub genes, among which MAPK1, AKT1, and TNF stood out due to their central position in the network. These genes regulate signaling and inflammatory pathways, which are the main platforms for communication within cells.

Hub genes being central in the complex network means that probably they are the main directors of the shared molecular mechanisms unveiled in this study. Besides being parts of several enriched pathways, their dominant presence in these pathways unmistakably highlights their substantial role at the interface between the gastrointestinal and neuropsychiatric systems.

5.8 Implications for the Gut–Brain Axis

The findings from this research strongly support the concept of gut–brain axis as a genetically and molecularly integrated system. While previous studies were mainly looking at the neurobiological and microbiota modes of communication, our study, on the other hand, points at host genetics as major players in mediating gut-brain interactions.

The newly identified common genes, signaling pathways, and coordinated expression patterns that we have revealed serve as an indication that constipation along with psychiatric disorders can be outcomes of molecular disruptions which overlap. This broad perspective of gastrointestinal-neuropsychiatric comorbidity has led to the opening of new paths for understanding and encourages further explorations.

CHAPTER 5

CONCLUSION

5. Conclusion

This article performs a bioinformatics-driven study of the genetic and molecular grounds shared by constipation, depression, and anxiety in their connection through the gut-brain axis. The analysis has revealed as many as 380 genes in common among the three disorders, which signifies a high level of genetic overlap and points to the existence of interconnected biological pathways.

The functional enrichment analysis showed that most of these genes participate in intracellular signaling, immune response, and neurotransmission-related activities. Among the main pathways identified as significantly enriched are the MAPK, PI3K-Akt, TNF signaling, and neuroactive ligand-receptor interaction pathways, which might play a role in gastrointestinal as well as neuropsychiatric functions.

The study of tissue-specific gene expression indicated that the common genes are expressed in both the brain and gut tissues. The positive correlation between brain and gut expression levels ($r = 0.553$, $p = 7.84e-32$) strongly indicates coordinated molecular activity between these two systems. Besides, the protein-protein interaction (PPI) network analysis highlighted hub genes such as MAPK1, AKT1, and TNF, indicating their pivotal roles in controlling the common signaling pathways.

The research results go hand in hand with the idea that constipation and mental disorders are not separate conditions but rather are connected through common genetic and molecular mechanisms. The study underlines the role of the host genetic background in gut-brain axis studies and delivers comprehension at the systems-level of gastro-intestinal-neuropsychiatric co-morbidities.

The above observations can pave the way for finding biomarkers and therapeutic targets for the combined treatment of these disorders. However, there is a need for further experimental studies to establish the functions of the identified genes and pathways.

CHAPTER 6

FUTURE PROSPECTS

6. Future Prospects

6.1 Experimental Validation of Findings

Since this work relies heavily on computational and database analyses, experimental validation of the candidate genes and pathways will be a very useful step. Going forward, transcriptomic (RNA-seq), proteomic, and functional experiments are recommended to establish the biological significance of these findings.

6.2 Establishing Causal Relationships

The paper points out shared genes and pathways but does not get into how these genes might cause different disease traits. To determine cause-and-effect, researchers can consider methods like Mendelian randomization, following people over time in cohort studies, and performing functional genomics experiments.

6.3 Cell-Type and Spatial Expression Analysis

Even though tissue-specific expression analysis demonstrated well-coordinated activity of genes in brain and gastrointestinal tissues, it will require cell-type level work and spatial transcriptomics to reveal the detailed biological context.

6.4 Integration of Multi-Omics Data

New studies should consider integrating data from epigenomics, metabolomics, and microbiome to present a more detailed picture of gut–brain interactions as well as the combined effects of genetic, environmental, and microbial factors.

6.5 Biomarker and Therapeutic Target Exploration

Experimental research and clinical trials need to be conducted for the hub genes discovered, such as MAPK1, AKT1, and TNF, to explore them as possible biomarkers and treatment targets.

6.6 Population-Based and Patient-Specific Studies

Since this study drew upon publicly available datasets, subsequent experiments should use data from individual patients along with population-based analyses so as to verify the results across varied demographic groups and enhance clinical relevance.

6.7 Expansion to Related Disorders

It would be beneficial to apply this method to other related conditions such as irritable bowel syndrome (IBS), neurodegenerative diseases, and additional mental health conditions, which might aid in uncovering a wider molecular underpinning of gut–brain axis disorders.

6.8 Dietary and Lifestyle Interventions

Investigations must delve into how dietary and lifestyle modifications can alter the gut–brain axis. Should a definitive mechanistic connection between diet, gut physiology, and neuropsychiatric outcomes be confirmed, these interventions may be employed as supplemental or supportive therapeutic modalities; nevertheless, additional experimental and clinical validation is warranted.

CHAPTER 7

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